PRE-ALGEBRA (7TH) · UNIT 1: ALGEBRAIC EXPRESSIONS AND INTEGERS

Deep Sea Expedition

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students pilot a research submarine. Depths, temperatures, and pressures become integer expressions, the dive log shows patterns, and sonar discoveries are mapped on the coordinate plane.

The vertical number line of the ocean gives meaning to adding, subtracting, and comparing integers.

Standards Addressed

- 7.NS.A.1 Add and subtract rational numbers; distance as absolute value of the difference
- 7.NS.A.2–3 Multiply and divide integers; solve problems
- 7.EE.A.1–2 Write and evaluate expressions; order of operations
- 6.NS.C.6–8 Coordinate plane review

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with an ocean-depth chart. Part 1 integer depths and changes.
Day 2	Part 2 dive expressions and order of operations.
Day 3	Part 3 sonar map and expedition log.

Materials

- Student handout and Discovery Cards
- Grid paper
- Calculators (CP1/Honors)
- Optional: ocean zone poster

How the Three Versions Differ

Level	What changes
CP2	Ordering depths, one-operation changes, a single-rate expression, plotting points, and simple patterns.
CP1	Multi-step integer computations, two-rate expressions, nested order of operations, and route depth changes.
Honors	Pressure and temperature models with limits, pattern rules and sums, grid distances, and reflections.

Watch For (Common Misconceptions)

- Thinking −860 > −120 because 860 > 120.
- Subtracting a negative incorrectly.
- Squaring before applying the negative sign incorrectly ($-3(6)^2$ vs $(-18)^2$).

Extension Ideas

- Research the depths of real ocean animals.
- Graph temperature vs. depth.
- Write a story problem for a classmate to solve.

Answer Key – CP2

Part 1: Depths & Integers

#	Task	Answer
1	Order the six depths from deepest to shallowest.	−1,250, −860, −310, −120, −70, −45
2	How many feet deeper is Squid Canyon than Octopus Reef?	−120 − (−860) = 740 ft
3	The sub goes from the Shipwreck (−310 ft) up 190 ft. What is its new depth?	−310 + 190 = −120 ft (Octopus Reef)
4	Find |−860| and explain what it means.	860; Squid Canyon is 860 ft from the surface.

Part 2: Dive Expressions

#	Task	Answer
5	Write an expression for the depth after m minutes.	−40m
6	Evaluate your expression for m = 3 and m = 8.	−120 ft and −320 ft
7	The water temperature drops 2°F for every 100 ft from 68°F. Find the temperature at −300 ft.	68 + 3(−2) = 62°F
8	Evaluate 68 − 2 × 3 + (−4) using the order of operations.	68 − 6 − 4 = 58

Part 3: Sonar Map

#	Task	Answer
9	Plot and label all six sonar points on a coordinate grid.	Points plotted at the given coordinates.
10	Which discoveries are in Quadrant IV? Quadrant II?	IV: Squid Canyon (4, −5), Vent Field (6, −6). II: Coral Garden (−5, 3), Kelp Forest (−1, 4).
11	The pattern of dive depths is −40, −80, −120, … What are the next three terms? What is the rule?	−160, −200, −240; subtract 40 each time.

Reflection

#	Prompt	Look for
R1	Why are depths below sea level negative?	Sea level is 0; below it is the opposite direction from above, so we use negative numbers.
R2	How do you subtract a negative integer?	Add its opposite: a − (−b) = a + b.

Answer Key – CP1

Part 1: Dive Data

#	Task	Answer
1	Find the change in depth from Coral Garden to Vent Field and from Vent Field back to the surface.	−1,250 − (−45) = −1,205 ft; 0 − (−1,250) = +1,250 ft
2	The average depth of the six discoveries: show the expression and evaluate it.	(−120 − 860 − 45 − 310 − 1,250 − 70) ÷ 6 = −2,655 ÷ 6 = −442.5 ft
3	The temperature at −860 ft is 41°F and at the surface it is 68°F. Find the average change per 100 ft (to the nearest tenth).	(41 − 68) ÷ 8.6 ≈ −3.1°F per 100 ft
4	Write an inequality comparing −310 and −120 and explain it in context.	−310 < −120; the Shipwreck is deeper than Octopus Reef.

Part 2: Expressions & Operations

#	Task	Answer
5	Write an expression for the depth after descending d minutes and then rising r minutes.	−40d + 60r
6	Evaluate your expression for d = 25, r = 10. Is the sub above or below Squid Canyon?	−1,000 + 600 = −400 ft; above Squid Canyon (−860).
7	Oxygen: the tank holds 900 units and the crew uses 12 units per minute. Write an expression for the oxygen left after t minutes and find how much is left after 45 minutes.	900 − 12t; 900 − 540 = 360 units
8	Evaluate the sonar reading $-3[4-(-2)]^2 \div 9 + 7$.	$-3(6)^2 \div 9 + 7 = -108 \div 9 + 7 = -12 + 7 = -5$

Part 3: Plan the Route

#	Task	Answer
9	Plot the points. Which two discoveries are in the same quadrant? Which discovery is closest to the origin (the dock)?	Squid Canyon and Vent Field are both in Quadrant IV. The Shipwreck (2, 1) is closest to the origin.
10	The sub visits Kelp Forest, then Coral Garden, then Octopus Reef. Find the total change in depth for the trip.	−70 → −45: +25; −45 → −120: −75; total −50 ft
11	Design a 4-stop dive route. Give each depth change as an integer and show that the sub returns to the surface (total change 0 from the start).	Answers vary; check that the changes sum to the opposite of the starting depth plus the return.

Reflection

#	Prompt	Look for
R1	Why does a rise add a positive number and a dive add a negative number?	Up is the positive direction from 0 and down is negative, so changes take those signs.
R2	How does writing an expression help you plan before you dive?	One expression gives the depth for any number of minutes, so you can test plans quickly.

Answer Key – Honors

Part 1: Pressure & Temperature

#	Task	Answer
1	Write an expression for the pressure at depth −d ft (d > 0). Find the pressure at Squid Canyon to the nearest tenth.	$1 + \frac{d}{33}$; $1 + \frac{860}{33} \approx 27.1$ atmospheres
2	Temperature falls 3°F per 100 ft from 68°F but never drops below 38°F. At what depth does it reach 38°F?	$68 - 3(\frac{d}{100}) = 38 \rightarrow d = 1{,}000 \rightarrow -1{,}000$ ft
3	Find the temperature at the Vent Field (−1,250 ft) using your rule. Why might the real vent water be much hotter?	Below −1,000 ft the rule gives 38°F; vents release heated water from inside the Earth.
4	Evaluate $\frac{-3(-5)^2+4(-6)}{-3}$ and explain each step.	−3(25) + (−24) = −75 − 24 = −99; −99 ÷ (−3) = 33

Part 2: Dive Patterns

#	Task	Answer
5	List the depths of dives 1–5 and write a rule for dive n.	−90, −120, −150, −180, −210; depth = −60 − 30n
6	Which dive first reaches Squid Canyon depth or deeper?	$-60 - 30n \le -860 \rightarrow n \ge 26.7 \rightarrow$ dive 27
7	The crew notices the sum of depths after n dives: −90, −210, −360, … Find the sum after 5 dives, then explain how the pattern grows.	−90 −120 −150 −180 −210 = −750; each sum adds a deeper (more negative) term.
8	Is it always true that the sum of two negative integers is less than each of them? Justify.	Yes; adding a negative moves left on the number line, so the sum is less than each addend.

Part 3: Route Analysis

#	Task	Answer
9	The sub travels from Coral Garden (−5, 3) to Vent Field (6, −6) moving only along grid lines. What is the shortest travel distance in feet?	$\lvert 6 - (-5) \rvert + \lvert -6 - 3 \rvert = 11 + 9 = 20$ units = 2,000 ft
10	Reflect the Shipwreck (2, 1) across the x-axis and then the y-axis. Where does it land? Which quadrant?	(−2, −1); Quadrant III
11	Design a route that visits four discoveries and returns to (0, 0). Find its grid distance and its total depth change and explain how you chose the order.	Answers vary; total depth change returning to the surface is 0.

Reflection

#	Prompt	Look for
R1	Explain why the temperature model needs a limit (38°F).	A linear rule would keep decreasing forever, but real water stops cooling; models have a valid domain.
R2	How did inductive reasoning help you predict which dive reaches Squid Canyon?	Finding the pattern rule let you solve for n instead of listing every dive.